Sahaj Bhadja

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Summary

Software engineer with 2+ years building and maintaining **multi-layer web applications, REST APIs, relational databases, and tested cloud services**, with strong ownership of quality, troubleshooting, usability, and reliable delivery.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Database Systems, Advanced NLP, Analysis of Algorithms, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Data Structures, Object-Oriented Programming, Operating Systems

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Built an Azure-backed multi-tenant application with **normalized relational schemas, row-level isolation, and indexed queries**, reducing dashboard latency from 1.2 seconds to 350 milliseconds for 500+ daily campus users.
- Reworked asynchronous FastAPI workflows with **unit-tested state transitions, failure isolation, debugging, metrics, and logs**, tripling throughput for 1K+ concurrent jobs while improving overall production reliability and maintainability.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Developed React and Flask features backed by **RESTful APIs and relational data models**, supporting 150+ instructors and 10K+ daily requests for uploads, grading history, model controls, and feedback.
- Implemented concurrent Python ingestion with **validation, automated testing, defect diagnosis, and user-facing error handling** for PDF, DOCX, ZIP, OCR, and multimodal workloads, reducing peak feedback turnaround 60%.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) *India*

- Designed cross-platform APIs and AWS data pipelines using **object-oriented architecture, Glue, S3, and SQL**, improving response times 30% while reducing technical debt 20% across critical business integrations.
- Automated 20+ unit and integration suites using PyTest, Mockito, Jenkins, and virtual machines, strengthening **interoperability, regression validation, defect resolution, and cross-platform application quality** across distributed customer services.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern - (VibrantChem) *India*

- Built eight modular services for authentication, resources, billing, and permissions, synchronizing 2M+ records through **REST APIs, WebSockets, relational schemas, reusable components, and responsive React interfaces** across locations.
- Automated Docker and Jenkins releases, reducing deployment time 50% while applying **Sentry monitoring, structured logging, issue diagnosis, and rollback-ready CI/CD practices** across eight customer-facing production service environments.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Translated customer workflows into technical specifications for 12+ integrations, mapping bookings, payments, coupons, and statuses into **reliable Flask APIs, database schemas, webhooks, and clearly documented application changes**.
- Built responsive interfaces and payment services processing 50K+ records with sub-200-millisecond responses, adding **input validation, authentication, defect handling, and integration testing** for accurate customer transactions across platforms.

Skills

Languages & Web: JavaScript, TypeScript, Python, Java, C#, React, Angular, Node.js, HTML5, CSS3, Responsive Design, Web Applications
Backend & Data: FastAPI, Flask, RESTful APIs, Webhooks, PostgreSQL, SQL Server, MySQL, Oracle, SQL, Relational Databases, Data Modeling, Multi-Layer Architecture
Testing & Quality: Unit Testing, Integration Testing, Regression Testing, Validation, PyTest, Mockito, Debugging, Defect Resolution, Performance Tuning, Code Reviews, Secure Development
Cloud & Delivery: AWS, Azure, GCP, Docker, Kubernetes, CI/CD, Jenkins, GitHub Actions, Git, Linux, Containers, Cloud-Native Development, Version Control
Engineering: Object-Oriented Programming, Data Structures, Algorithms, Application Maintenance, Requirements Analysis, Technical Documentation, Agile, AI-Assisted Development, Machine Learning, User-Centered Design

Projects

Military Housing Operations Portal | *React, TypeScript, C#, ASP.NET Core, SQL Server*

- Designed a **multi-layer military housing application** for unit allocation, waitlists, leases, resident payments, and maintenance requests using responsive React workflows, RESTful APIs, role-based access, and normalized relational schemas.

Maintenance Workflow Quality Suite | *C#, xUnit, PostgreSQL, Docker, GitHub Actions*

- Built an **automated quality and defect-diagnosis suite** validating work-order APIs, database changes, permissions, interoperability, and failure paths through unit and integration tests, containerized environments, CI checks, and structured logs.

Chat2SQL | *Java, TypeScript, SQL, Docker, Maven, Node.js*

- Extended a **cross-platform database workspace** supporting 10+ relational systems through Java and TypeScript, adding schema visualization, query execution, Docker-based environments, and documented workflows for repeatable testing and troubleshooting.